

MCP IN .NET

FROM ZERO TO



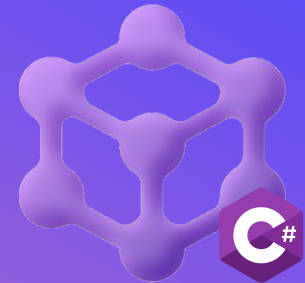


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isolutions[®]



MOTIVATION



one of
played
at risks,
cut into

dictatorial /ˈdɪktəˈtɔːriəl/ *adv.*
like a dictator. 2 overbearing.
orally *adv.* [Latin: related
TATOR]

diction /ˈdɪkʃ(ə)n/ *n.* manner of
ciation in speaking or singing
dictio from *dico dict-* say]

dictionary /ˈdɪkʃənəri/ *n.* (pl
book listing (usu. alphabetical)
explaining the words of a language
giving corresponding words in
language. 2 reference book
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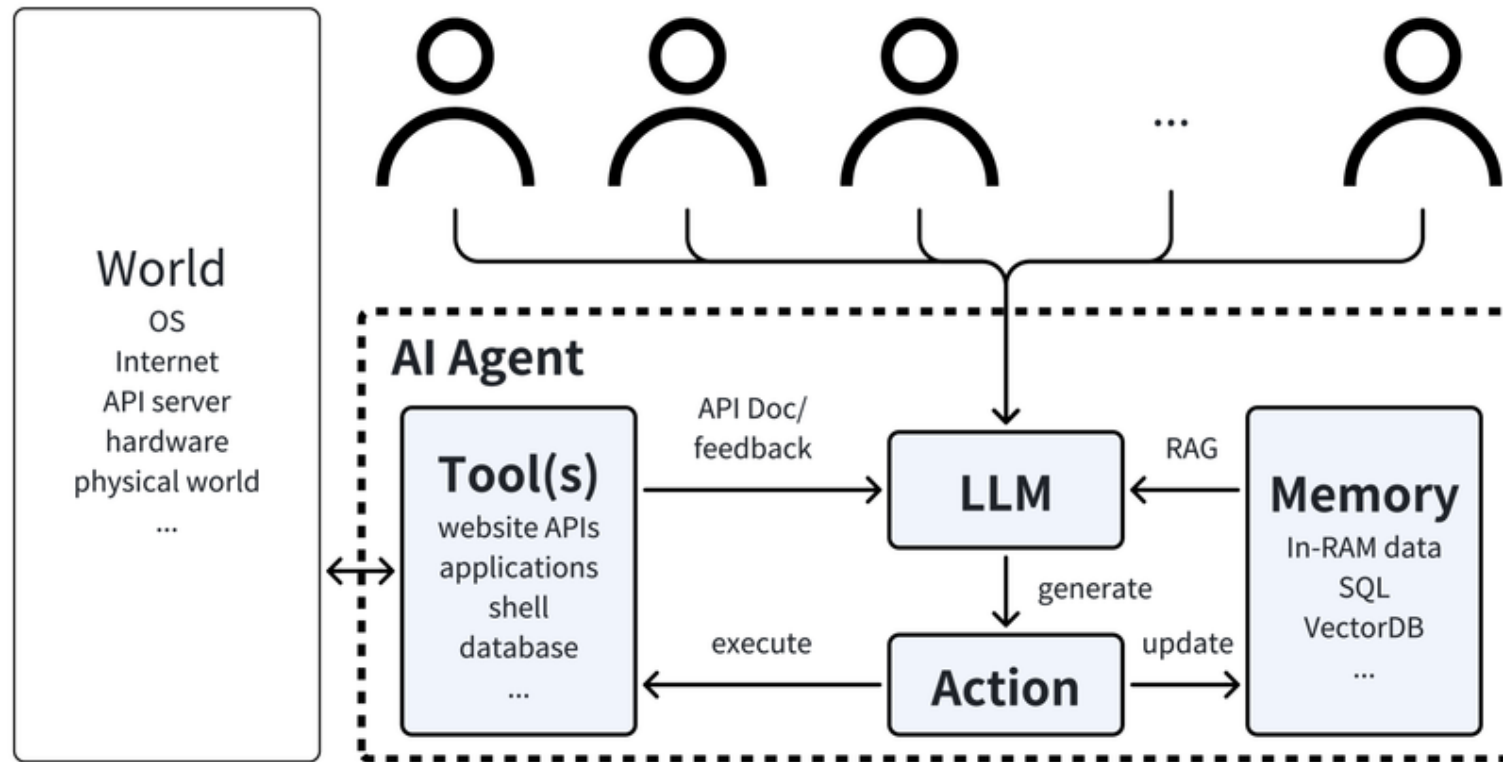
What is the current date?

1113



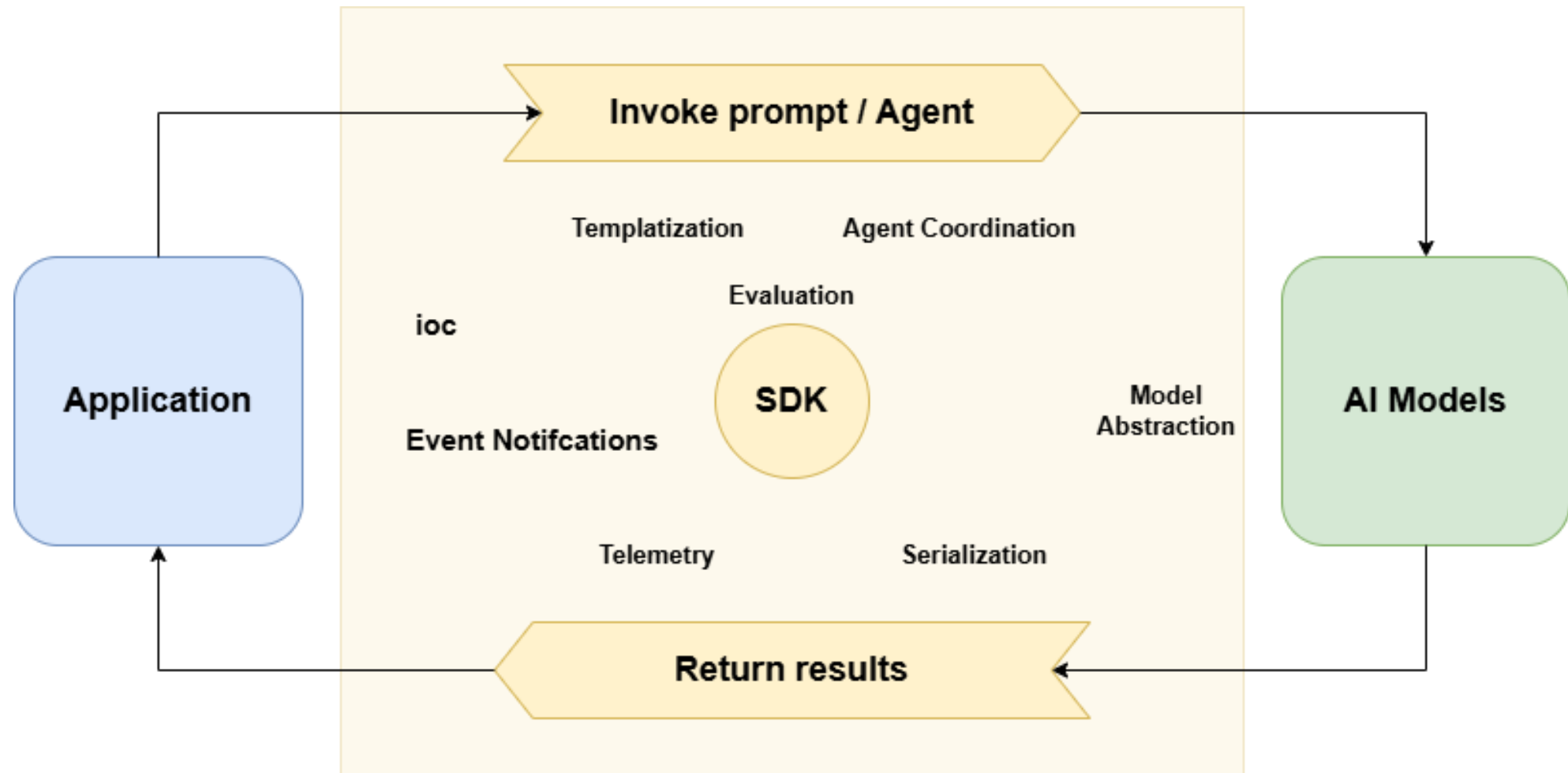


LLM BASED AI AGENT



C# ❤️ AI

WHY WE NEED AN SDK



C# SDKS

Microsoft.Extensions.AI (Middleware)

Microsoft.Extensions.AI.Abstractions

Microsoft.Extensions.VectorData

AI SDKS
(OllamaSharp, OpenAI, Azure AI Inference)

Vector Store SDKs

Agent Libraries
(Semantic Kernel, AutoGen, Agent Framework)

AgentEval / MEAI.Evaluation

C# SDKS



Semantic Kernel

+



Autogen

=



Microsoft Agent
Framework



ModelContextProtocol
ModelContextProtocol.Core
ModelContextProtocol.AspNetCore

MEAI – SETUP

```
1 var endpoint = "";
2 var apiKey = "";
3 var modelId = "";
4
5 var azureClient = new AzureOpenAIClient(
6     new Uri(endpoint),
7     new ApiKeyCredential(apiKey));
8
9 var chatClient = azureClient.GetChatClient(modelId).AsIChatClient();
```

MEAI – EXECUTE PROMPT

```
1 // initialize history & settings
2 var prompt = "How do I get from Basel to Mannheim?";
3 List<ChatMessage> history = [new ChatMessage(ChatRole.User, prompt)];
4 var options = new ChatOptions();
5
6 // execute prompt
7 var response = await chatClient.GetResponseAsync(history, options);
8 //response.Text
9
```

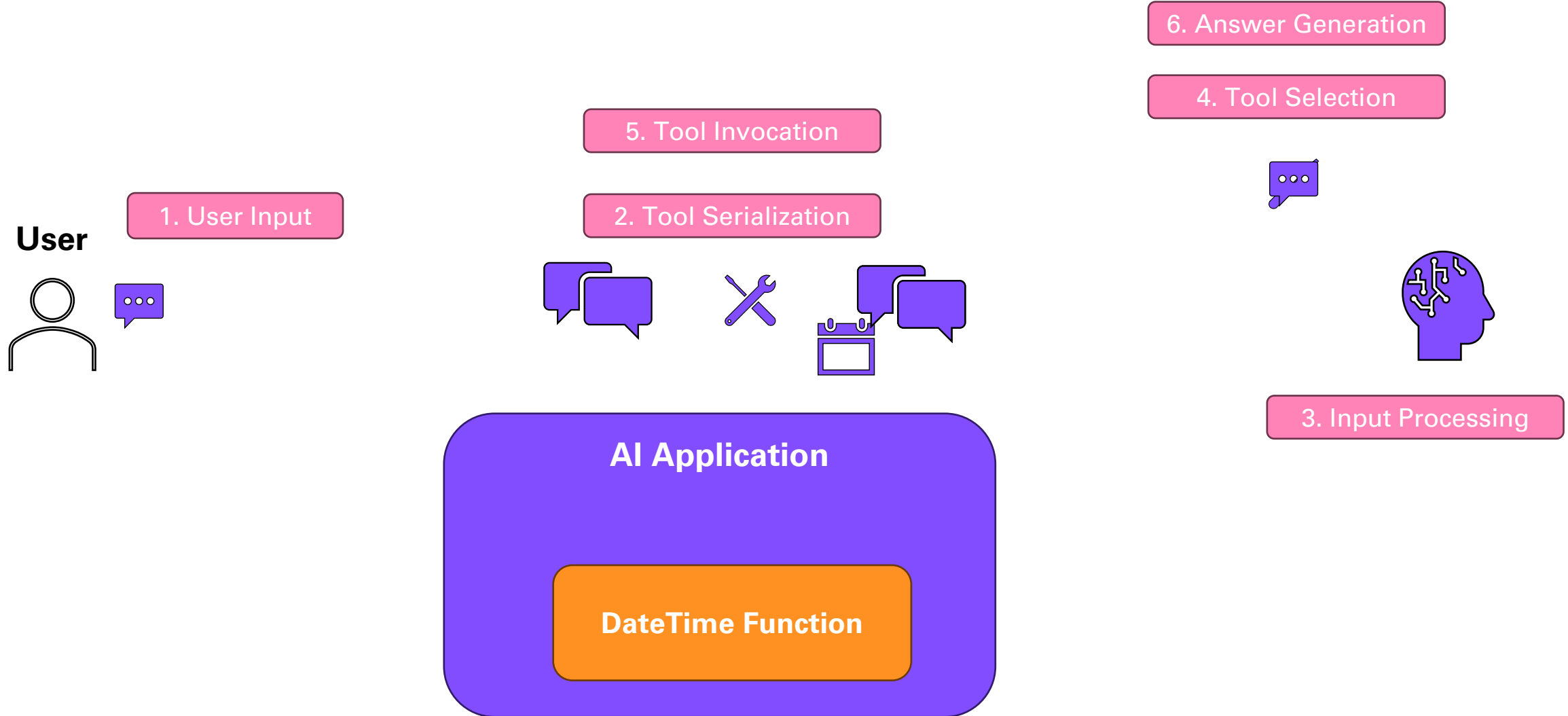


What is the current date?

FUNCTION CALLING



FUNCTION CALLING



MEAI – FUNCTION CALLING

```
1 var functionCallClient = new ChatClientBuilder(chatClient)
2   .UseFunctionInvocation()
3   .Build();
4
5 var options= new ChatOptions
6 {
7     Tools = [AIFunctionFactory.Create(
8         () => DateTime.UtcNow,
9         "get_current_datetime",
10        "Gets the current UTC date and time")],
11     ToolMode = ChatToolMode.Auto // RequireAny, None
12 };
13
14 var response = await functionCallClient.GetResponseAsync(history, options);
```

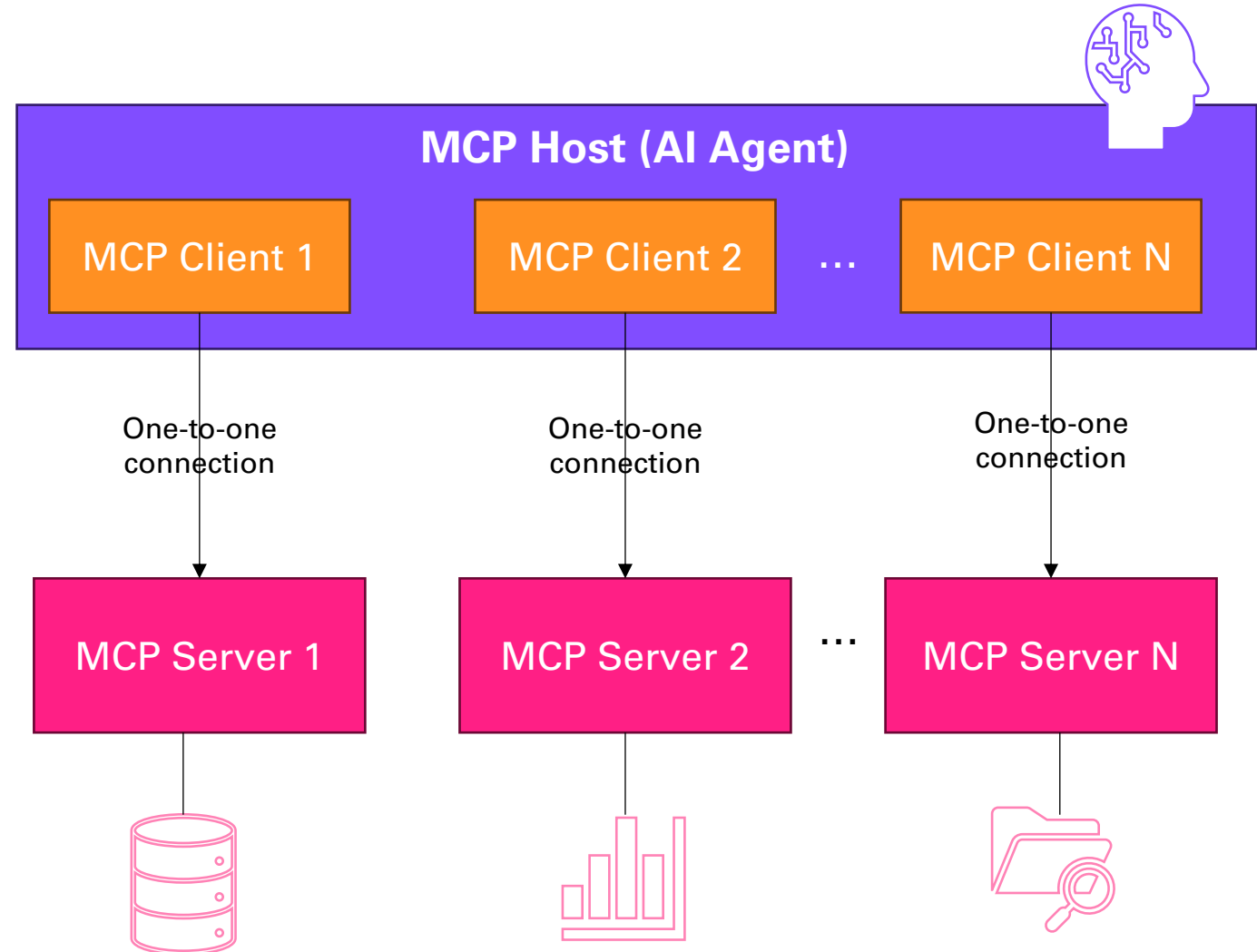
DEMO



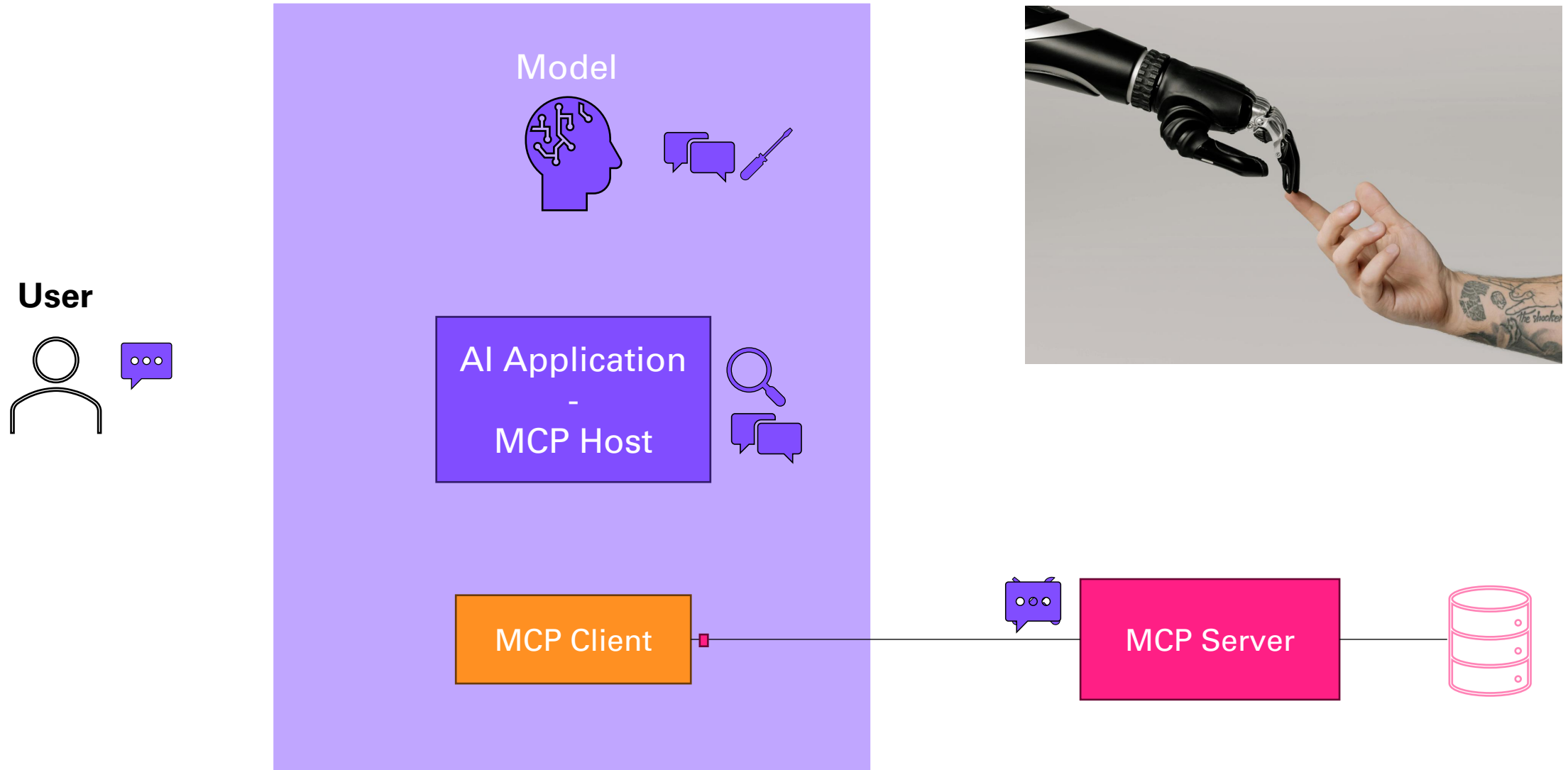
MCP IN .NET

MODEL CONTEXT PROTOCOL (MCP)

- Client – Server
- Transport
 - Stdio (local)
 - Http (remotely)
 - Server-Sent Events for Streaming
- JSON-RPC



MCP - FLOW



MCP SERVER - TOOL

```
1  [McpServerToolType]
2  public class DateTools
3  {
4      [McpServerTool]
5      [Description("Returns the current date and time in ISO 8601 format.")]
6      public DateTime GetCurrentDateTime(IMcpServer mcpServer)
7          ⇒ DateTime.UtcNow;
8  }
9
```

MCP SERVER

```
1  var builder = WebApplication.CreateBuilder(args);  
2  
3  builder.Services  
4      .AddMcpServer()  
5      .WithHttpTransport()  
6      .WithTools<DateTools>();  
7  
8  // Add services to the container.  
9  var app = builder.Build();  
10  
11 // map the mcp endpoint  
12 app.MapMcp("/mcp");  
13  
14 await app.RunAsync();
```

AI AGENT & MCP CLIENT

```
1 // create mcp client
2 var transport = new HttpClientTransport(new HttpClientTransportOptions()
3 {
4     Endpoint = new Uri("https://localhost:7133/mcp"),
5     Name = "MCP Desktop Client",
6     TransportMode = HttpTransportMode.StreamableHttp,
7 }, httpClient);
8
9 await using IMcpClient mcpClient = await McpClient.CreateAsync(transport);
10
11 // Retrieve the list of tools available on the MCP server
12 var mcpTools = await mcpClient.GetMcpToolsAsAIFunctionsAsync();
13
14 // add tools to chatOptions and run execute prompt
15 ...
16
```

MCP CAPABILITIES

MCP Server

- Tools
- Resources
- Prompts
- Utilities
 - Completion
 - ~~Logging~~
 - Pagination

MCP Client

- ~~Roots~~
- Elicitation
- ~~Sampling~~

Extensions

- MCP Apps
- MCP Auth

<https://modelcontextprotocol.io>

MCP Draft 2026-07-28

DEMO



MCP INSPECTOR

```
npx @modelcontextprotocol/inspector
```

MCP Inspector v0.16.8

Transport Type

Streamable HTTP

URL

https://mcpoathsecurity-hag0drcke

Server Entry

Servers File

> Authentication

> Configuration

Reconnect

Disconnect

● Connected

Resources

Prompts

Tools

Ping

Sampling

Elicitations

Roots

Auth

Tools

List Tools

Clear

get_random_number

Generates a random number between the specified minimum and maximum values.

get_current_date_time

Returns the current date and time in ISO 8601 format.

get_random_number_from_date_time

Generates a random number based on a date.

get_current_date_time

Returns the current date and time in ISO 8601 format.

Run Tool

Copy Input

Tool Result: Success

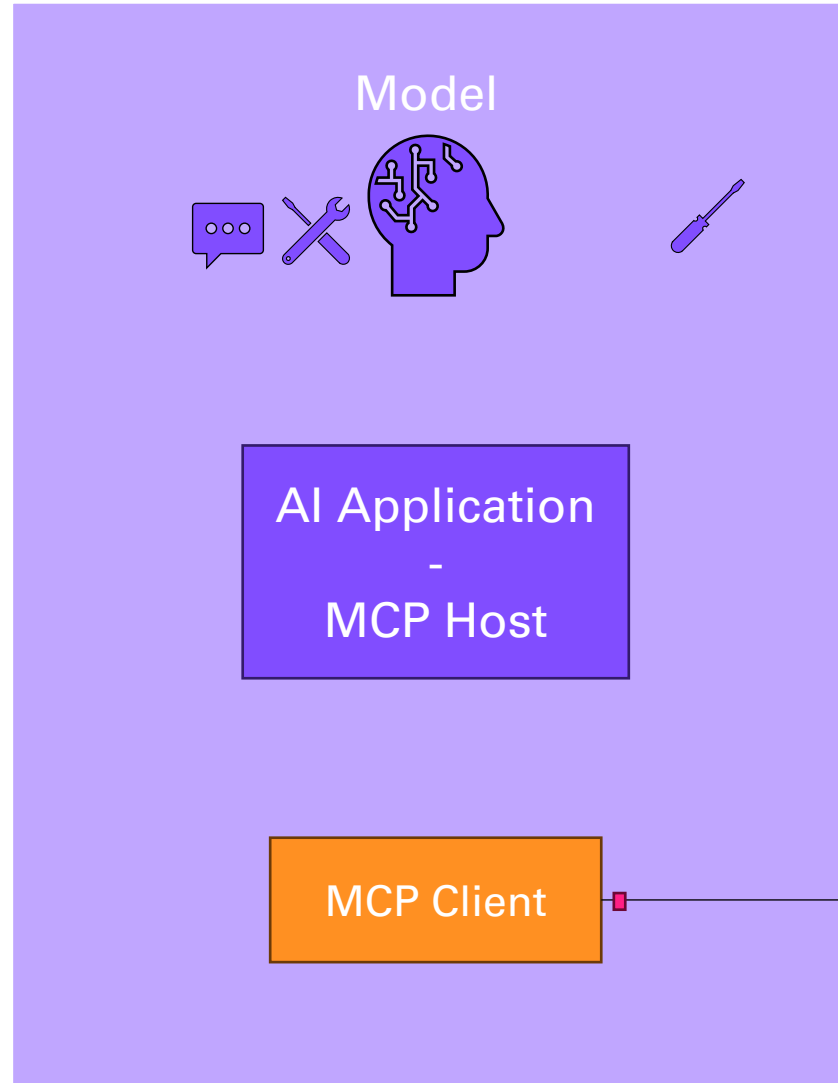
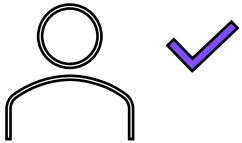
"2025-09-22T11:42:44.1786351Z"

HUMAN-IN-THE-LOOP



MCP - ELICITATION

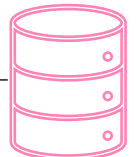
User



Human in the loop



MCP Server



MCP SERVER – ELICITATION

```
1 [McpServerTool]
2 [Description("Returns the current date and time in ISO 8601 format.")]
3 public Task<string> GetCurrentDateTime(IMcpServer mcpServer)
4 {
5     if (mcpServer.ClientCapabilities?.Elicitation == null)
6     {
7         throw new McpException("Client does not support elicitation");
8     }
9
10    // start elicitation
11    var userAccepted = await mcpServer.ElicitAsync(new ElicitRequestParams()
12    {
13        Message = $"Do you want to execute tool '{nameof(GetCurrentDateTime)}'",
14        RequestedSchema = new RequestSchema()
15        {
16            Properties = { ["answer"] = new BooleanSchema() }
17        }
18    });
19
20    if (userAccepted.Action != "accept" ||
21        userAccepted.Content?["answer"].ValueKind != JsonValueKind.True)
22    {
23        throw new McpException("User declined to proceed");
24    }
25
26    return DateTime.UtcNow.ToString("o");
27 }
```

MCP CLIENT – ELICITATION

```
1 var transport = ...
2 await using IMcpClient mcpClient = await McpClientFactory.CreateAsync(
3     transport,
4     new McpClientOptions()
5     {
6         ClientInfo = new()
7         {
8             Name = "McpDemoClient",
9             Version = "1.0.0",
10        },
11        Capabilities = new()
12        {
13            Elicitation = new()
14            {
15                Form = new(),
16            },
17        },
18        Handlers = new()
19        {
20            ElicitationHandler = HandleElicitationAsync,
21        }
22    }
23 );
```

MCP CLIENT – ELICITATION HANDLER

```
1 public static ValueTask<ElicitResult> HandleElicitationAsync(  
2     ElicitRequestParams? requestParams,  
3     CancellationToken token)  
4 {  
5     Console.WriteLine(requestParams.Message);  
6     Console.WriteLine($"Please allow/decline function execution with [y,n]");  
7     var key = Console.ReadKey();  
8     var userAllowance = key.KeyChar == 'y' || key.KeyChar == 'Y';  
9     var content = new Dictionary<string, JsonElement>();  
10    content["answer"] = JsonSerializer.Deserialize<JsonElement>(  
11        JsonSerializer.Serialize(userAllowance));  
12  
13    return ValueTask.FromResult(new ElicitResult  
14    {  
15        Action = "accept",  
16        Content = content  
17    });  
18 }
```

DEMO



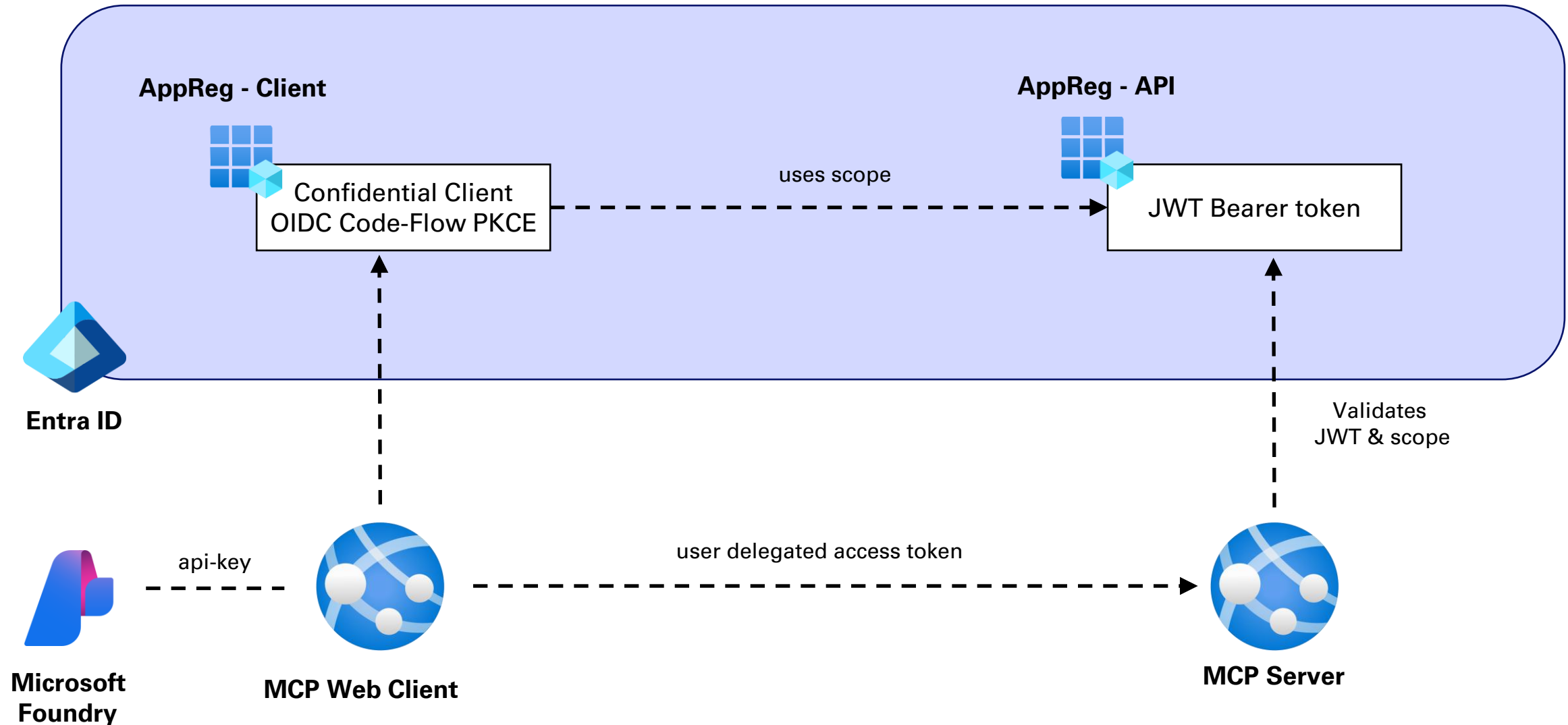
SECURE MCP SERVERS

SECURITY BEST PRACTICES

- MCP Server is basically an API – apply best-practices
- Client: Use OpenID Connect (OIDC)
 - Acquire the access token
- Server: Use delegated user access tokens only
 - Validate bearer token

The S in MCP stands for Security

DEMO SETUP - MICROSOFT ENTRA ID



MCP SERVER OAUTH SECURITY

```
1 var serverUrl = "";
2 var tenantId = "";
3 var authority = $"https://login.microsoftonline.com/{tenantId}/v2.0";
4
5 builder.Services.AddMicrosoftIdentityWebApiAuthentication(builder.Configuration);
6 builder.Services.AddAuthentication()
7     .AddMcp(options =>
8     {
9         options.ResourceMetadata = new()
10         {
11             AuthorizationServers = [authority],
12             ResourceName = "HttpMcpServer demo server",
13             Resource = new Uri(serverUrl ),
14             ResourceDocumentation = $"{{serverUrl }}/health",
15             ScopesSupported = ["mcp:tools"],
16         };
17     });
18
19 builder.Services.AddAuthorization();
20
```

MCP SERVER OAUTH SECURITY

```
1 // change to scp or scope if not using magic namespaces from MS
2 // The scope must be validate as we want to force only delegated access tokens
3 // The scope is requires to only allow access tokens intended for this API
4 builder.Services.AddAuthorizationBuilder()
5     .AddPolicy("mcp_tools", policy =>
6         policy.RequireClaim("http://schemas.microsoft.com/identity/claims/scope",
7             "mcp:tools"));
```

MCP SERVER OAUTH SECURITY

```
1 app.UseAuthentication();  
2 app.UseAuthorization();  
3  
4 app.MapMcp("/mcp").RequireAuthorization("mcp_tools");
```

CONFIDENTIAL OIDC MCP CLIENT

```
1 builder.Services.AddAuthentication(OpenIdConnectDefaults.AuthenticationScheme)
2   .AddMicrosoftIdentityWebApp(builder.Configuration.GetSection("AzureAd"))
3   .EnableTokenAcquisitionToCallDownstreamApi(
4     ["api://96b0f495-3b65-4c8f-a0c6-c3767c3365ed/mcp:tools"])
5   .AddInMemoryTokenCaches();
6
7 builder.Services.AddAuthorization(options =>
8 {
9   options.FallbackPolicy = options.DefaultPolicy;
10 });
11
12 builder.Services.AddRazorPages()
13   .AddMicrosoftIdentityUI();
```

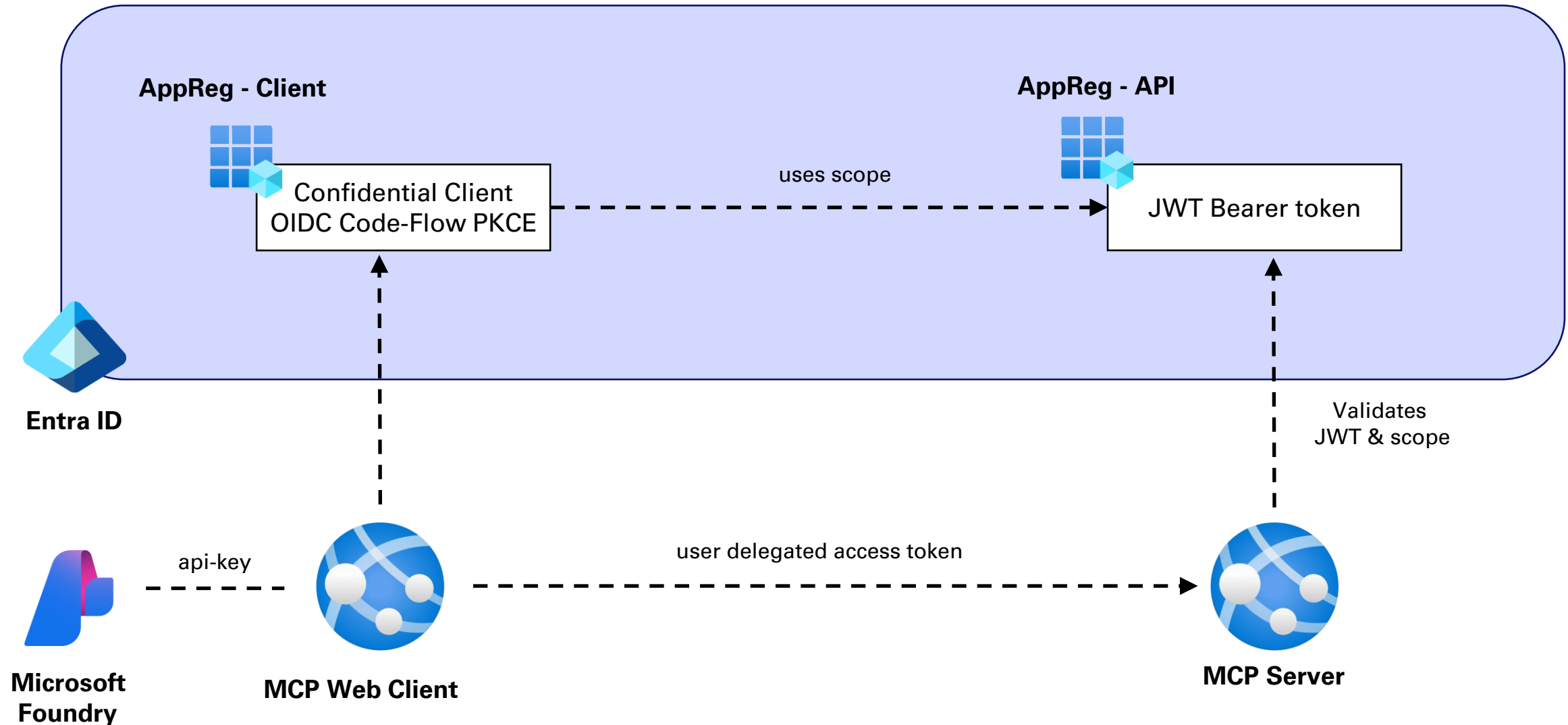
CONFIDENTIAL OIDC MCP CLIENT

```
1 var accessToken = await _tokenAcquisition
2   .GetAccessTokenForUserAsync(["api://96b0f495-3b65-4c8f-a0c6-c3767c3365ed/mcp:tools"]);
3
4 await _chatService.EnsureSetupAsync(_clientFactory, accessToken);
```

DEMO



DEMO SETUP - MICROSOFT ENTRA ID





RECAP & RECOMMENDATIONS

LINKS



[GitHub Demo](#)



[LinkedIn](#)



[Blog \(Medium\)](#)

REFERENCES

- [MEAI Documentation](#)
- [MCP C# SDK](#)
- [MCP Docs](#)
- [MCP Inspector](#)
- [MCP for beginners course](#)